

SEAT No. _____



No. of Printed Pages: 2

C51J

SARDAR PATEL UNIVERSITY

F.Y.B.sc. – CA & IT (SEM-I) Examination February – 2022

PS01CIT23 : Computer Organization and Digital Electronics

Date: 12-02-2022, Saturday

Time: 12-00 to 02-00 PM

Total Marks : 70

Q.1 Multiple choice questions:

[10]

1. The ALU and CU jointly known as
 - A. ALU
 - B. CPU
 - C. Memory
 - D. None
2. The computer size was very large in _____.
 - A. First generation
 - B. Third generation
 - C. Second generation
 - D. Fourth generation
3. Hexadecimal number E is equal to decimal number
 - A. 13
 - B. 16
 - C. 14
 - D. 15
4. Binary number system base _____.
 - A. 8
 - B. 2
 - C. 7
 - D. 9
5. The ____ gate has two or more input signals. All inputs must be low to get a low output.
 - A. NAND
 - B. NOR
 - C. AND
 - D. OR
6. De Morgan's second theorem says that a NAND gate is equivalent to a _____.
 - A. bubbled OR
 - B. bubbled NOR
 - C. bubbled AND
 - D. None
7. Invert gate has only _____ input and _____ output.
 - A. Two, One
 - B. Two, Two
 - C. One, One
 - D. One, Two
8. A _____ is a memory element that stores a binary digit.
 - A. Decoder
 - B. Multiplexer
 - C. Flip flop
 - D. Encoder
9. In full adder XOR gate's output is _____.
 - A. Carry
 - B. Sum
 - C. Reminder
 - D. None
10. In shift right register, the arrival of the first rising clock edge sets the _____ flip-flop
 - A. left
 - B. right
 - C. up
 - D. down

Q.2 (A) Fill In the blanks.

[08]

1. CPU stands for _____.
2. Octal number system base _____.
3. The _____ gate has two or more input signals. All inputs must not be same to get a high output.
4. A multiplexer also called a _____.

C 1J

[P.T.O.]

(B) TRUE/FALSE

1. CPU is the brain of any computer systems.
2. $37_{(10)} = 101111_{(2)}$
3. A combinational circuit that performs the arithmetic addition of three bits is called Full adder.
4. Boolean Algebra and K-MAP both are way to simplify the equation.

Q.3 Attempt any *Ten* out of *Twelve*.

[20]

1. Differentiate Primary Memory and Secondary Memory.
2. Differentiate – Hardware and Software.
3. Explain floppy disk, cd rom and pen drive in brief.
4. What is number system? How many type of number system.
5. $101011_{(2)} = ?_{(16)}$
6. $254_{(8)} = ?_{(2)}$
7. Explain distributive law.
8. Explain De Morgan first theorem.
9. Explain half adder in brief.
10. What is RS Flip flop?
11. What is multiplexer?
12. Define shift register.

Q.4 Attempt any *Four* out of *Eight*.

[32]

1. Draw the Block diagram of Computer and explain generation of it.
2. What is printer and explain types of printer.
3. Explain Decimal to Octal, Binary conversion with example.
4. Explain HexaDecimal to Binary and Octal conversion with example.
5. Explain full adder in detail.
6. Explain NAND, OR, XNOR, NOT gate.
7. Explain encoder in detail.
8. Explain comparator detail.

————— X —————